TRIAL-RISK EARLY-WARNING · ADVANCED ENGINE

Optimal Transport + nested Active Inference

A continuous, anticipatory layer on top of the threshold rules — it surfaces risk **earlier**, is **geometry- & tail-aware**, and **tells you what to check next**.

OPTIMAL TRANSPORT
Wasserstein drift

How far has the distribution drifted from a meta-learned reference?

Wasserstein distance respects clinical severity — it sees mass move toward the dangerous tail (Hy's-Law territory) where a mean or a single threshold is silent.

Flags drift before a line is crossed — and the transport map says which analyte/grade is driving it.

ACTIVE INFERENCE
free energy + EFE

- **A model of each subject's expected trajectory.**

Surprise (free energy) rising = early warning — the subject is doing something the model didn't expect, before a frank event.

Expected Free Energy picks the most informative next assessment — e.g. an unscheduled LFT — and recommends it for approval.

NESTED + META
one deep hierarchy

Participant → study → client as one model — priors flow down, prediction-errors (risk) flow up.

Priors are meta-learned across studies (a Wasserstein barycenter of prior cohorts) — a new study starts population-informed.

A participant signal propagates to study & program risk — not three siloed dashboards.

WHAT IT BUYS OVER THRESHOLDS ALONE

Earlier | Geometry- & tail-aware | Anticipatory | Coherent across tiers

Decision-support, advanced / research-grade. Validated & frozen before any regulated use. The rule-based DLT / Hy's-Law / QTcF / QTL triggers stay authoritative; validated tools own every number; the SRC / DSMB / medical monitor decide. EFE only recommends ***what to observe*** — for human approval, never an autonomous clinical or dosing action.

Companion to the Trial-Termination Early-Warning Dashboard and the Participant-Tier Advanced-Engine implementation spec. · Methods (OT + active inference) independently math-checked. · Illustrative.